

Designer's™ Data Sheet

TMOS E-FET™

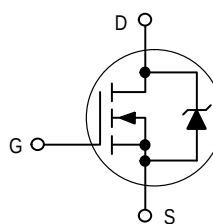
Power Field Effect Transistor

TO-247 with Isolated Mounting Hole

N-Channel Enhancement-Mode Silicon Gate

This high voltage MOSFET uses an advanced termination scheme to provide enhanced voltage-blocking capability without degrading performance over time. In addition, this advanced TMOS E-FET is designed to withstand high energy in the avalanche and commutation modes. The new energy efficient design also offers a drain-to-source diode with a fast recovery time. Designed for high voltage, high speed switching applications in power supplies, converters and PWM motor controls, these devices are particularly well suited for bridge circuits where diode speed and commutating safe operating areas are critical and offer additional safety margin against unexpected voltage transients.

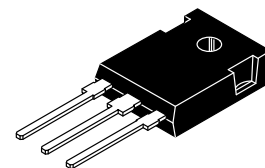
- Robust High Voltage Termination
- Avalanche Energy Specified
- Source-to-Drain Diode Recovery Time Comparable to a Discrete Fast Recovery Diode
- Diode is Characterized for Use in Bridge Circuits
- I_{DSS} and $V_{DS(on)}$ Specified at Elevated Temperature
- Isolated Mounting Hole Reduces Mounting Hardware



MTW20N50E

Motorola Preferred Device

TMOS POWER FET
20 AMPERES
500 VOLTS
 $R_{DS(on)} = 0.24 \text{ OHM}$



CASE 340K-01, Style 1
TO-247AE

MAXIMUM RATINGS ($T_C = 25^\circ\text{C}$ unless otherwise noted)

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DSS}	500	Vdc
Drain-Gate Voltage ($R_{GS} = 1.0 \text{ M}\Omega$)	V_{DGR}	500	Vdc
Gate-Source Voltage — Continuous	V_{GS}	± 20	Vdc
— Non-Repetitive ($t_p \leq 10 \text{ ms}$)	V_{GSM}	± 40	Vpk
Drain Current — Continuous	I_D	20	Adc
— Continuous @ 100°C	I_D	14.1	Apk
— Single Pulse ($t_p \leq 10 \mu\text{s}$)	I_{DM}	60	Apk
Total Power Dissipation Derate above 25°C	P_D	250 2.0	Watts W/ $^\circ\text{C}$
Operating and Storage Temperature Range	T_J, T_{stg}	-55 to 150	$^\circ\text{C}$
Single Pulse Drain-to-Source Avalanche Energy — Starting $T_J = 25^\circ\text{C}$ ($V_{DD} = 100 \text{ Vdc}$, $V_{GS} = 10 \text{ Vdc}$, $I_L = 20 \text{ Apk}$, $L = 10 \text{ mH}$, $R_G = 25 \Omega$)	E_{AS}	2000	mJ
Thermal Resistance — Junction to Case	$R_{\theta JC}$	0.50	$^\circ\text{C/W}$
— Junction to Ambient	$R_{\theta JA}$	40	$^\circ\text{C/W}$
Maximum Lead Temperature for Soldering Purposes, 1/8" from case for 10 seconds	T_L	260	$^\circ\text{C}$

Designer's Data for "Worst Case" Conditions — The Designer's Data Sheet permits the design of most circuits entirely from the information presented. SOA Limit curves — representing boundaries on device characteristics — are given to facilitate "worst case" design.

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Preferred devices are Motorola recommended choices for future use and best overall value.

MTW20N50E**ELECTRICAL CHARACTERISTICS** ($T_J = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit	
OFF CHARACTERISTICS						
Drain–Source Breakdown Voltage (V _{GS} = 0 Vdc, I _D = 250 μAdc) Temperature Coefficient (Positive)	V _{(BR)DSS}	500 —	— 583	— —	Vdc mV/°C	
Zero Gate Voltage Drain Current (V _{DS} = 500 Vdc, V _{GS} = 0 Vdc) (V _{DS} = 500 Vdc, V _{GS} = 0 Vdc, T _J = 125°C)	I _{DSS}	— —	— —	10 100	μAdc	
Gate–Body Leakage Current (V _{GS} = ± 20 Vdc, V _{DS} = 0)	I _{GSS}	—	—	100	nAdc	
ON CHARACTERISTICS (1)						
Gate Threshold Voltage (V _{DS} = V _{GS} , I _D = 250 μAdc) Temperature Coefficient (Negative)	V _{GS(th)}	2.0 —	3.0 7.0	4.0 —	Vdc mV/°C	
Static Drain–Source On–Resistance (V _{GS} = 10 Vdc, I _D = 10 Adc)	R _{DS(on)}	—	0.20	0.24	Ohm	
Drain–Source On–Voltage (V _{GS} = 10 Vdc) (I _D = 20 Adc) (I _D = 10 Adc, T _J = 125°C)	V _{DS(on)}	— —	5.75 —	6.0 6.0	Vdc	
Forward Transconductance (V _{DS} = 15 Vdc, I _D = 10 Adc)	g _{FS}	11	16.2	—	mhos	
DYNAMIC CHARACTERISTICS						
Input Capacitance	(V _{DS} = 25 Vdc, V _{GS} = 0 Vdc, f = 1.0 MHz)	C _{iss}	—	3880	pF	
Output Capacitance		C _{oss}	—	452		
Reverse Transfer Capacitance		C _{rss}	—	96		
SWITCHING CHARACTERISTICS (2)						
Turn–On Delay Time	(V _{DD} = 250 Vdc, I _D = 20 Adc, V _{GS} = 10 Vdc, R _G = 9.1 Ω)	t _{d(on)}	—	29	ns	
Rise Time		t _r	—	90		
Turn–Off Delay Time		t _{d(off)}	—	97		
Fall Time		t _f	—	84		
Gate Charge (See Figure 8)	(V _{DS} = 400 Vdc, I _D = 20 Adc, V _{GS} = 10 Vdc)	Q _T	—	100	nC	
		Q ₁	—	20		
		Q ₂	—	44		
		Q ₃	—	36		
SOURCE–DRAIN DIODE CHARACTERISTICS						
Forward On–Voltage (1)	(I _S = 20 Adc, V _{GS} = 0 Vdc) (I _S = 20 Adc, V _{GS} = 0 Vdc, T _J = 125°C)	V _{SD}	— —	0.916 0.81	1.1 —	Vdc
Reverse Recovery Time (See Figure 14)	(I _S = 20 Adc, V _{GS} = 0 Vdc, dI _S /dt = 100 A/μs)	t _{rr}	—	431	ns	
		t _a	—	272		
		t _b	—	159		
Reverse Recovery Stored Charge			Q _{RR}	—	6.67	—
INTERNAL PACKAGE INDUCTANCE						
Internal Drain Inductance (Measured from the drain lead 0.25″ from package to center of die)	L _D	—	5.0	—	nH	
Internal Source Inductance (Measured from the source lead 0.25″ from package to source bond pad)	L _S	—	13	—	nH	

(1) Pulse Test: Pulse Width $\leq 300\text{ }\mu\text{s}$, Duty Cycle $\leq 2\%$.

(2) Switching characteristics are independent of operating junction temperature.

TYPICAL ELECTRICAL CHARACTERISTICS

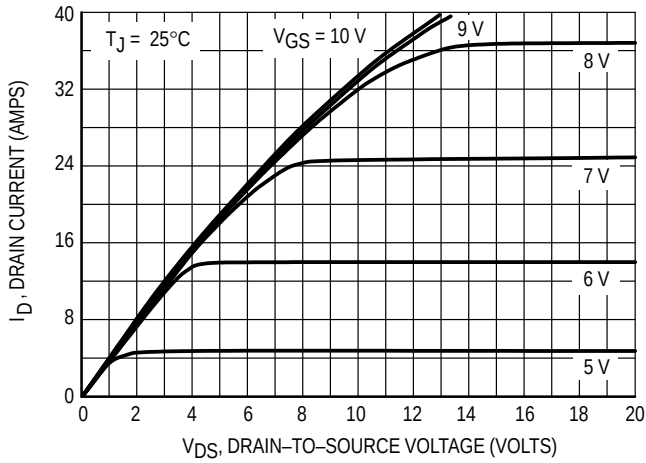


Figure 1. On-Region Characteristics

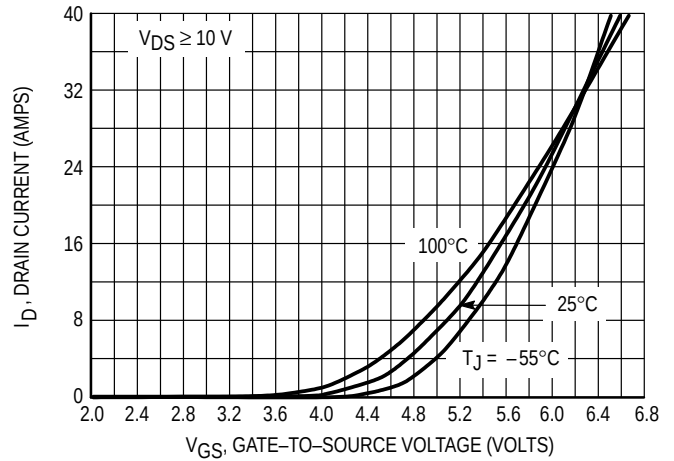


Figure 2. Transfer Characteristics

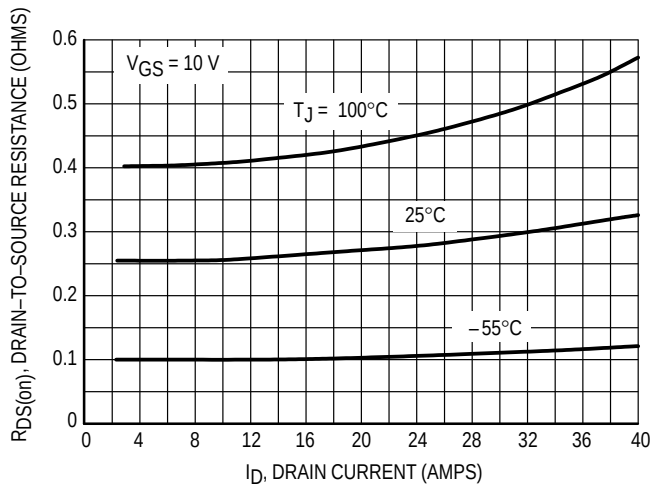


Figure 3. On-Resistance versus Drain Current and Temperature

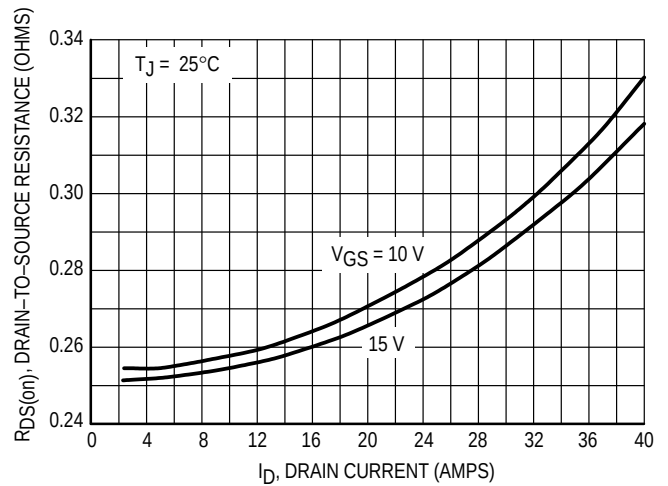


Figure 4. On-Resistance versus Drain Current and Gate Voltage

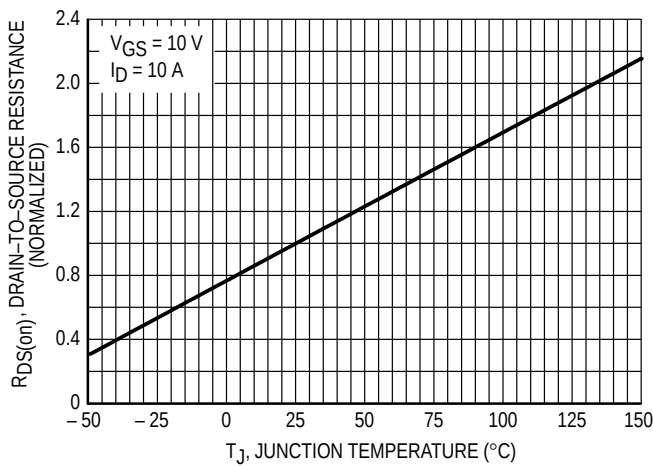


Figure 5. On-Resistance Variation with Temperature

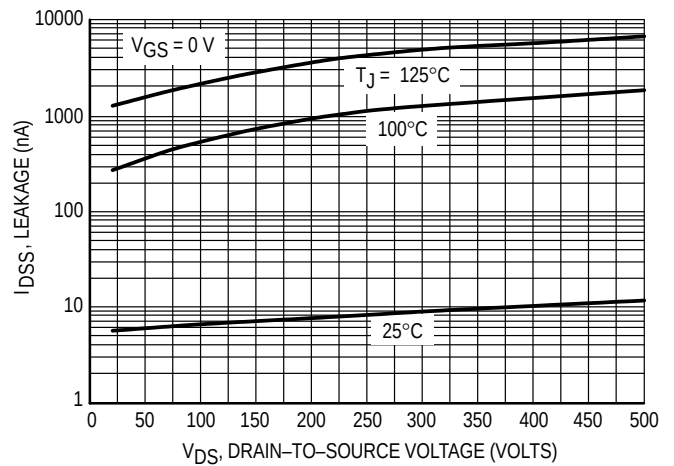


Figure 6. Drain-To-Source Leakage Current versus Voltage

POWER MOSFET SWITCHING

Switching behavior is most easily modeled and predicted by recognizing that the power MOSFET is charge controlled. The lengths of various switching intervals (Δt) are determined by how fast the FET input capacitance can be charged by current from the generator.

The published capacitance data is difficult to use for calculating rise and fall because drain–gate capacitance varies greatly with applied voltage. Accordingly, gate charge data is used. In most cases, a satisfactory estimate of average input current ($I_{G(AV)}$) can be made from a rudimentary analysis of the drive circuit so that

$$t = Q/I_{G(AV)}$$

During the rise and fall time interval when switching a resistive load, V_{GS} remains virtually constant at a level known as the plateau voltage, V_{GSP} . Therefore, rise and fall times may be approximated by the following:

$$t_r = Q_2 \times R_G / (V_{GG} - V_{GSP})$$

$$t_f = Q_2 \times R_G / V_{GSP}$$

where

V_{GG} = the gate drive voltage, which varies from zero to V_{GG}

R_G = the gate drive resistance

and Q_2 and V_{GSP} are read from the gate charge curve.

During the turn–on and turn–off delay times, gate current is not constant. The simplest calculation uses appropriate values from the capacitance curves in a standard equation for voltage change in an RC network. The equations are:

$$t_{d(on)} = R_G C_{iss} \ln [V_{GG}/(V_{GG} - V_{GSP})]$$

$$t_{d(off)} = R_G C_{iss} \ln (V_{GG}/V_{GSP})$$

The capacitance (C_{iss}) is read from the capacitance curve at a voltage corresponding to the off–state condition when calculating $t_{d(on)}$ and is read at a voltage corresponding to the on–state when calculating $t_{d(off)}$.

At high switching speeds, parasitic circuit elements complicate the analysis. The inductance of the MOSFET source lead, inside the package and in the circuit wiring which is common to both the drain and gate current paths, produces a voltage at the source which reduces the gate drive current. The voltage is determined by $L di/dt$, but since di/dt is a function of drain current, the mathematical solution is complex. The MOSFET output capacitance also complicates the mathematics. And finally, MOSFETs have finite internal gate resistance which effectively adds to the resistance of the driving source, but the internal resistance is difficult to measure and, consequently, is not specified.

The resistive switching time variation versus gate resistance (Figure 9) shows how typical switching performance is affected by the parasitic circuit elements. If the parasitics were not present, the slope of the curves would maintain a value of unity regardless of the switching speed. The circuit used to obtain the data is constructed to minimize common inductance in the drain and gate circuit loops and is believed readily achievable with board mounted components. Most power electronic loads are inductive; the data in the figure is taken with a resistive load, which approximates an optimally snubbed inductive load. Power MOSFETs may be safely operated into an inductive load; however, snubbing reduces switching losses.

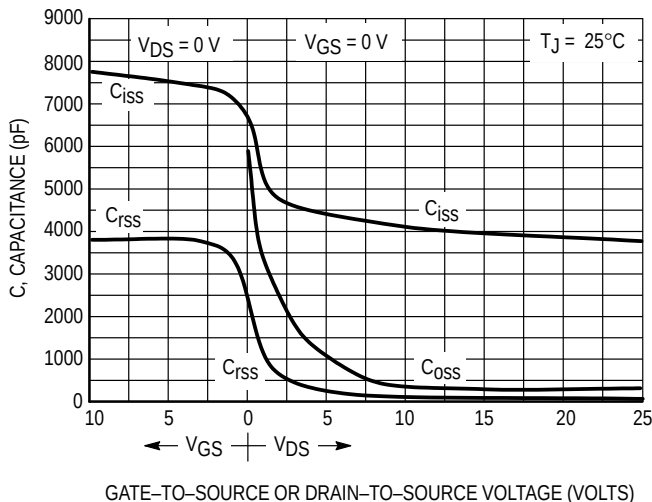


Figure 7a. Capacitance Variation

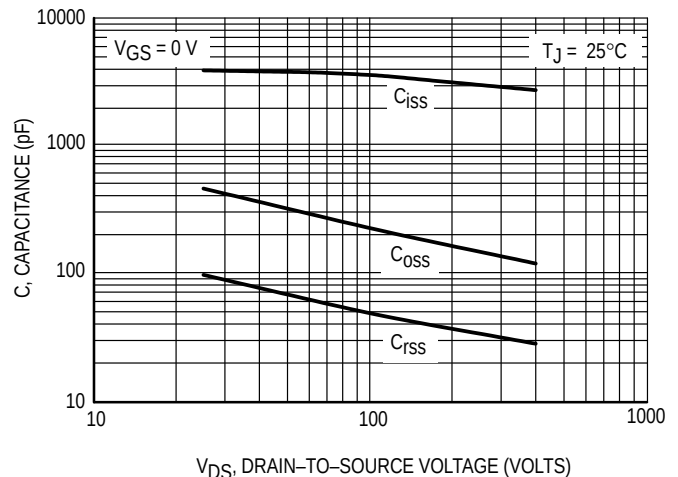


Figure 7b. High Voltage Capacitance Variation

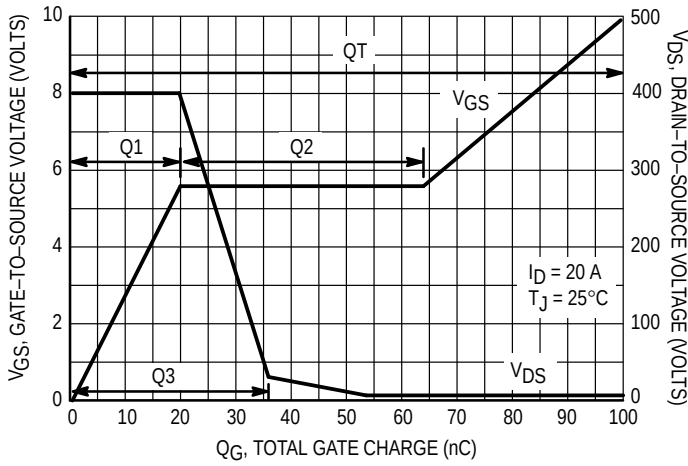


Figure 8. Gate-To-Source and Drain-To-Source Voltage versus Total Charge

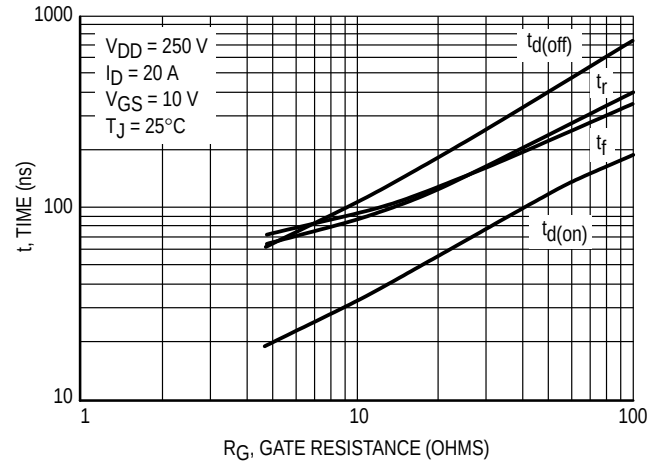


Figure 9. Resistive Switching Time Variation versus Gate Resistance

DRAIN-TO-SOURCE DIODE CHARACTERISTICS

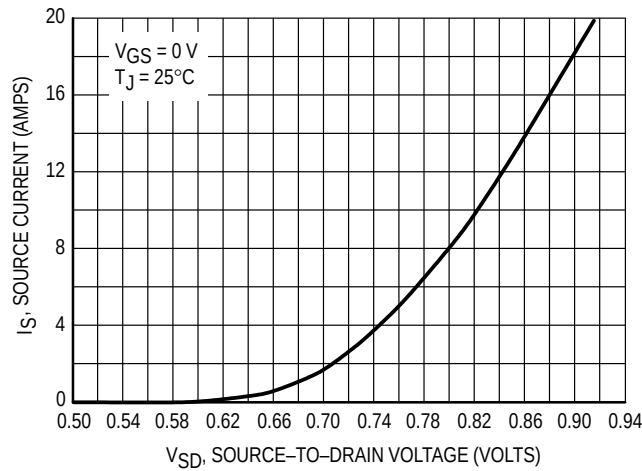


Figure 10. Diode Forward Voltage versus Current

SAFE OPERATING AREA

The Forward Biased Safe Operating Area curves define the maximum simultaneous drain-to-source voltage and drain current that a transistor can handle safely when it is forward biased. Curves are based upon maximum peak junction temperature and a case temperature (T_C) of 25°C. Peak repetitive pulsed power limits are determined by using the thermal response data in conjunction with the procedures discussed in AN569, "Transient Thermal Resistance—General Data and Its Use."

Switching between the off-state and the on-state may traverse any load line provided neither rated peak current (I_{DM}) nor rated voltage (V_{DSS}) is exceeded and the transition time (t_r, t_f) do not exceed 10 μs . In addition the total power averaged over a complete switching cycle must not exceed $(T_J(MAX) - T_C)/(R_{\theta JC})$.

A Power MOSFET designated E-FET can be safely used in switching circuits with unclamped inductive loads. For reli-

able operation, the stored energy from circuit inductance dissipated in the transistor while in avalanche must be less than the rated limit and adjusted for operating conditions differing from those specified. Although industry practice is to rate in terms of energy, avalanche energy capability is not a constant. The energy rating decreases non-linearly with an increase of peak current in avalanche and peak junction temperature.

Although many E-FETs can withstand the stress of drain-to-source avalanche at currents up to rated pulsed current (I_{DM}), the energy rating is specified at rated continuous current (I_D), in accordance with industry custom. The energy rating must be derated for temperature as shown in the accompanying graph (Figure 12). Maximum energy at currents below rated continuous I_D can safely be assumed to equal the values indicated.

SAFE OPERATING AREA

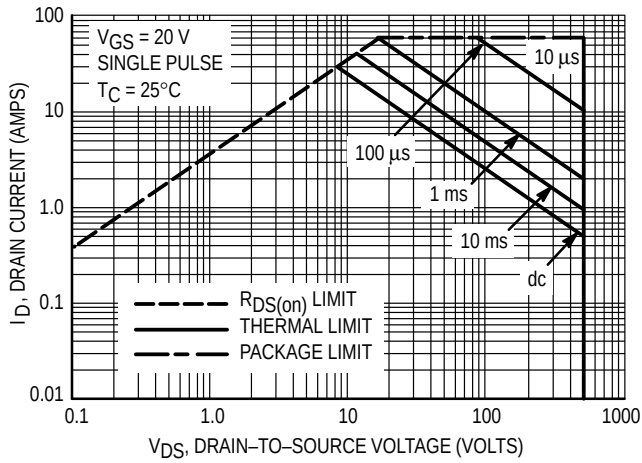


Figure 11. Maximum Rated Forward Biased Safe Operating Area

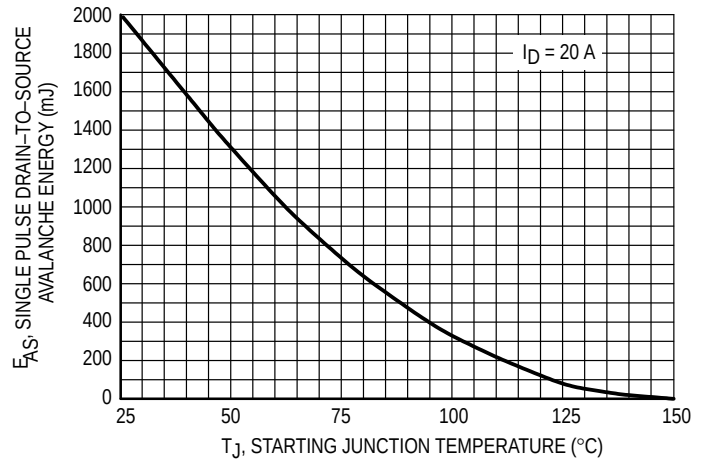


Figure 12. Maximum Avalanche Energy versus Starting Junction Temperature

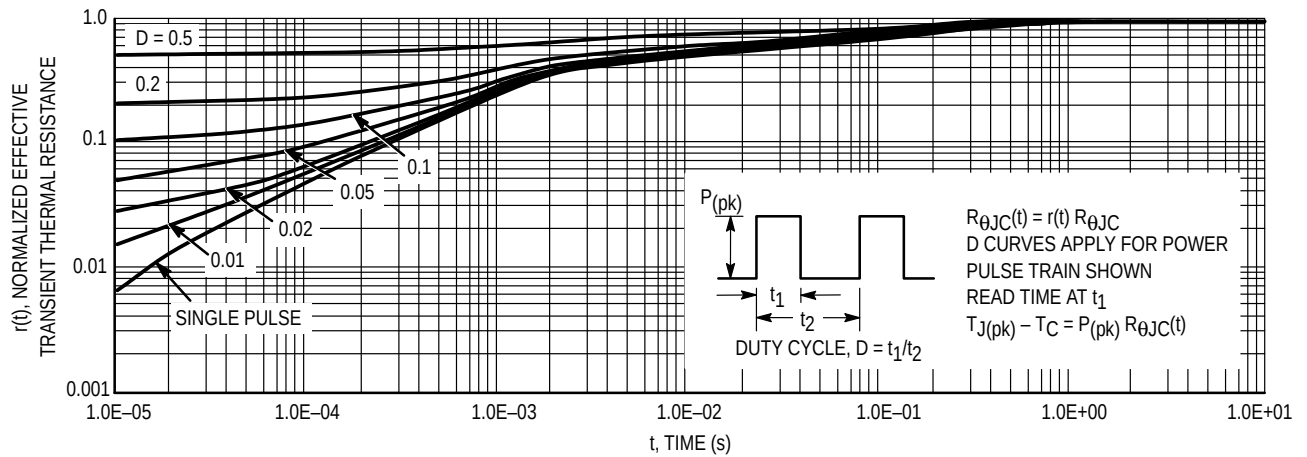


Figure 13. Thermal Response

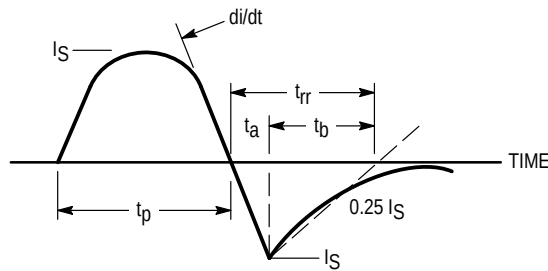
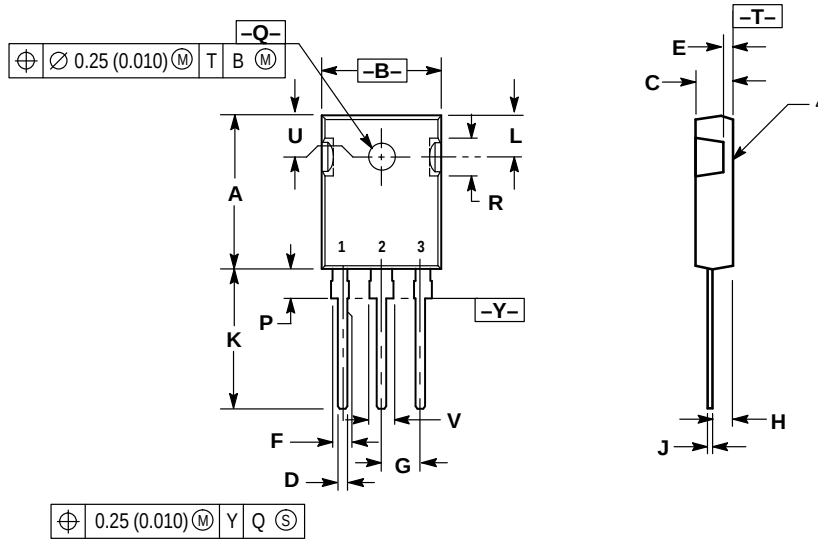


Figure 14. Diode Reverse Recovery Waveform

PACKAGE DIMENSIONS



- NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
 2. CONTROLLING DIMENSION: MILLIMETER.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	19.7	20.3	0.776	0.799
B	15.3	15.9	0.602	0.626
C	4.7	5.3	0.185	0.209
D	1.0	1.4	0.039	0.055
E	1.27 REF		0.050 REF	
F	2.0	2.4	0.079	0.094
G	5.5 BSC		0.216 BSC	
H	2.2	2.6	0.087	0.102
J	0.4	0.8	0.016	0.031
K	14.2	14.8	0.559	0.583
L	5.5 NOM		0.217 NOM	
P	3.7	4.3	0.146	0.169
Q	3.55	3.65	0.140	0.144
R	5.0 NOM		0.197 NOM	
U	5.5 BSC		0.217 BSC	
V	3.0	3.4	0.118	0.134

STYLE 1:
 PIN 1. GATE
 2. DRAIN
 3. SOURCE
 4. DRAIN

**CASE 340K-01
 ISSUE O**

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